

IB Mathematics AA HL

Integration by Parts — MCQ Practice

Exam-style multiple-choice questions

Instructions: Select the best answer for each question.

1. Evaluate $\int x e^x dx$.

- A. $xe^x + C$
- B. $e^x(x - 1) + C$
- C. $e^x(x + 1) + C$
- D. $x^2e^x + C$

2. For $\int x \ln x dx$, the most appropriate choice for u is

- A. x
- B. $\ln x$
- C. $x \ln x$
- D. $1/x$

3. Evaluate $\int x \cos x dx$.

- A. $x \sin x + \cos x + C$
- B. $x \sin x - \cos x + C$
- C. $x \cos x + \sin x + C$
- D. $x \cos x - \sin x + C$

4. $\int \ln x dx =$

- A. $x \ln x + C$
- B. $x \ln x - x + C$
- C. $\ln x - x + C$
- D. $x/\ln x + C$

5. Which integral requires integration by parts twice?

- A. $\int e^x dx$
- B. $\int x e^x dx$
- C. $\int x^2 e^x dx$
- D. $\int \sin x dx$

6. Evaluate $\int x^2 e^x dx$.

- A. $e^x(x^2 - 2x + 2) + C$
- B. $e^x(x^2 + 2x + 2) + C$

C. $x^2 e^x + C$

D. $e^x(x^2 - x + 1) + C$

7. Evaluate $\int_0^1 x e^x dx$.

A. $e - 1$

B. 1

C. $1/e$

D. e

8. Evaluate $\int x \sin(2x) dx$.

A. $-(x/2)\cos(2x) + (1/4)\sin(2x) + C$

B. $(x/2)\cos(2x) + (1/4)\sin(2x) + C$

C. $-(x/2)\sin(2x) + (1/4)\cos(2x) + C$

D. $x \cos(2x) + \sin(2x) + C$

9. Given $I = \int_0^1 x e^{ax} dx$, $a \neq 0$, which is I?

A. $[e^a(a - 1) + 1]/a^2$

B. $[e^a(a + 1) - 1]/a^2$

C. $(e^a - 1)/a$

D. $(ae^a - 1)/a^2$

10. Evaluate $\int_0^{\pi/2} x \cos x dx$.

A. 1

B. $\pi/2 - 1$

C. $\pi/2 + 1$

D. 0

11. Evaluate $\int x^2 \sin x dx$.

A. $-x^2 \cos x + 2x \sin x + 2 \cos x + C$

B. $x^2 \cos x - 2x \sin x - 2 \cos x + C$

C. $-x^2 \sin x + 2x \cos x + 2 \sin x + C$

D. $x^2 \cos x + 2x \sin x + 2 \cos x + C$

12. If $I_n = \int x^n e^x dx$, which relationship is correct?

A. $I_n = x^n e^x + n I_{n-1}$

B. $I_n = x^n e^x - n I_{n-1}$

C. $I_n = n e^x - x^n I_{n-1}$

D. $I_n = x^n e^x + I_{n-1}$

Answer Key

Question	Answer	Question	Answer
1	B	2	B
3	A	4	B
5	C	6	A
7	B	8	A
9	A	10	A
11	A	12	B

Topics tested: choice of u and dv , repeated integration by parts, definite integrals, parameterized integrals and reduction relationships.